

Tool and AI Disclosure Document

Zeladija AI Customer Support Agent , CSAI 422: Applied Data Mining, Spring 2026

The table below discloses every external tool, library, model, dataset, and AI assistant used in the Zeladija AI Customer Support Agent, with justification for each choice and a description of how it fits into the overall architecture. All data used is synthetic or openly licensed. No real customer data or proprietary information was used.

Tool / Library	Category	Justification	Role in Architecture
OpenAI API (gpt-4o-mini)	LLM Provider	Cost-efficient GPT-4 class model suitable for classification and response generation within academic budget constraints.	Handles intent classification, response generation, fact extraction, escalation decisions, and LLM-as-judge evaluation. Receives conversation messages and retrieved policy text.
LangGraph (LangChain)	Orchestration	Provides a stateful, graph-based multi-agent framework ideal for modeling branching customer support workflows.	Defines the multi-agent StateGraph, conditional edge routing, and node transitions across all agent types.
LangChain Core	Agent Framework	Offers lightweight tool decorators and agent state abstractions that integrate cleanly with LangGraph.	Provides @tool decorators for mock API functions and manages agent state throughout the conversation lifecycle.
ChromaDB	Vector Database	Lightweight, open-source vector store with no external dependencies, ideal for local RAG pipelines.	Stores and retrieves document chunk embeddings for semantic search during the retrieval-augmented generation step.
all-MiniLM-L6-v2 (sentence-transformers)	Embedding Model	Fast, compact model with strong semantic performance, runs locally with no data leaving the system.	Converts document chunks and user queries into vector representations used for ChromaDB similarity search.
unitary/toxic-bert (HuggingFace)	Classification Model	Pre-trained toxicity classifier enabling guardrail enforcement before routing messages to the LLM.	Classifies incoming customer messages for toxicity; triggers content guardrails if thresholds are exceeded.
rank_bm25	Retrieval Library	Lightweight BM25 implementation enabling keyword-based retrieval to complement dense vector search in a hybrid setup.	Computes keyword match scores merged with embedding similarity via Reciprocal Rank Fusion for improved retrieval.
SQLite (Python stdlib)	Database	Zero-dependency relational store suitable for persisting structured session data across multi-turn conversations.	Stores customer facts extracted from conversations, enabling long-term memory across sessions.
Faker	Data Generation	Rapidly generates realistic synthetic order data without requiring real customer records.	Produces 200 synthetic customer orders (seed=42) used in development, testing, and evaluation.
Streamlit	Dashboard	Rapid prototyping framework enabling evaluation metrics to be visualized with minimal frontend code.	Renders the evaluation metrics dashboard displaying resolution rates, latency distributions, and guardrail results.
Plotly / Pandas	Visualization	Industry-standard Python libraries for data manipulation and interactive charting, well-integrated with Streamlit.	Powers all charts and data tables within the evaluation dashboard.
Chainlit	Frontend	Purpose-built chat UI framework with native LangGraph integration, reducing frontend boilerplate significantly.	Provides the conversational chat interface that connects user input to the LangGraph backend.
Bitext Customer Support Dataset (HuggingFace)	Dataset	Openly licensed dataset of customer support intents used to validate FAQ coverage and expand test case diversity.	Used locally for test case design and FAQ validation; not transmitted to any external API.
Claude (Anthropic)	AI Assistant	Used during development for boilerplate drafting, docstring formatting, and prompt template iteration.	Assisted in writing LangGraph codebase, system prompt templates, and code documentation. All output was reviewed, tested, and modified by us.